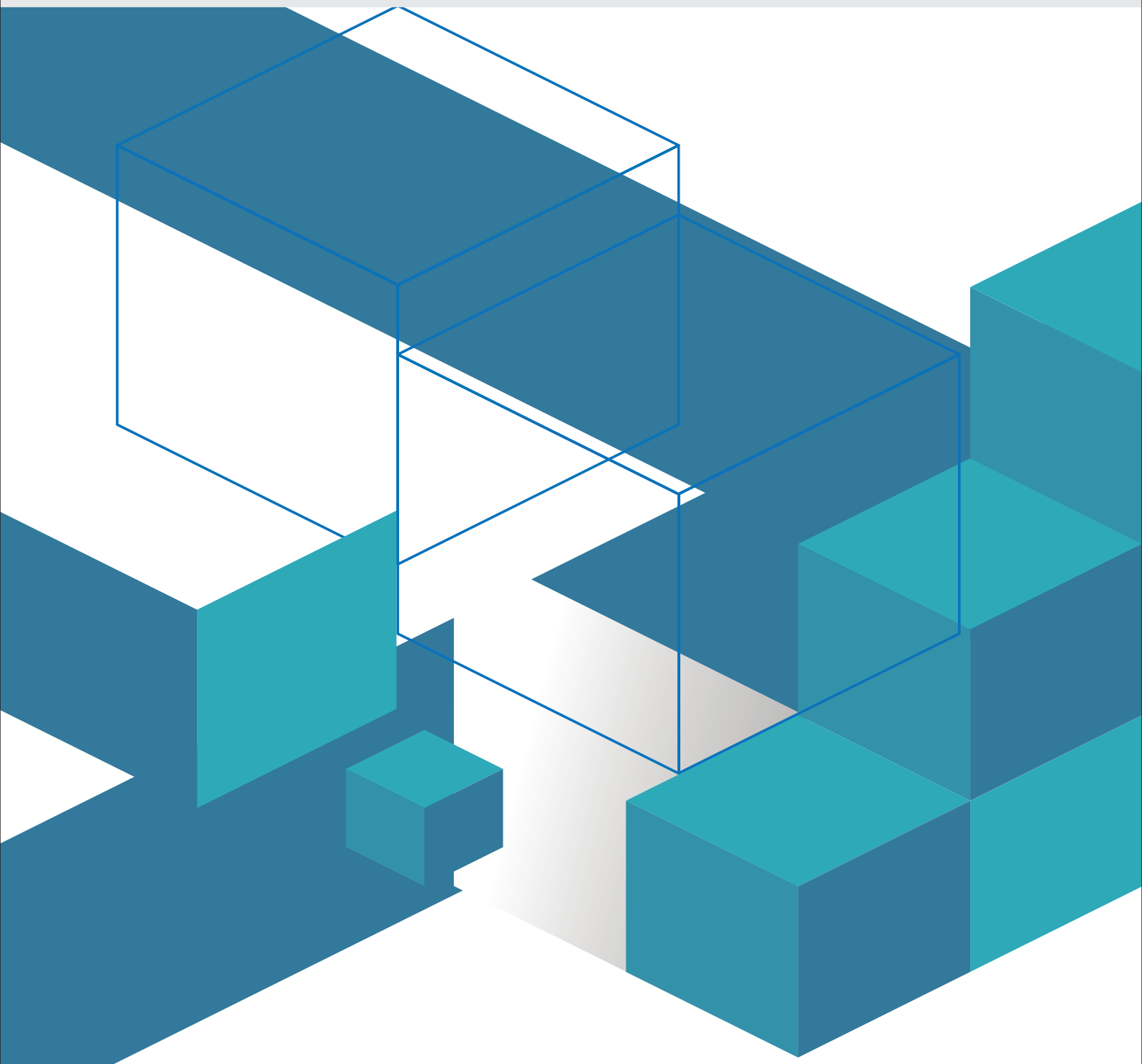
MATHEMATICS

INDEFINITE & DEFINITE INTEGRATION_MEGA_XII



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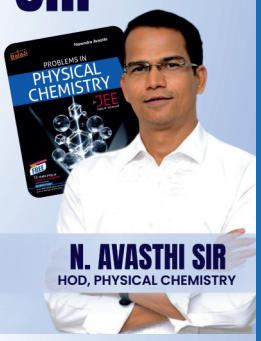

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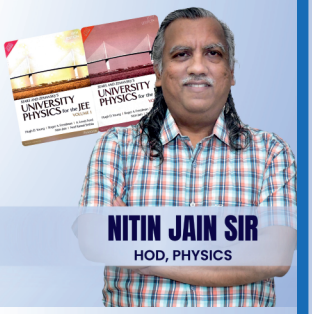


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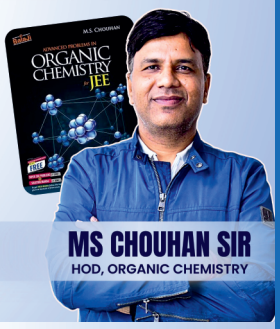


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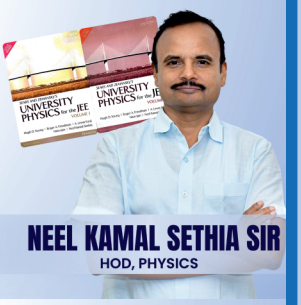


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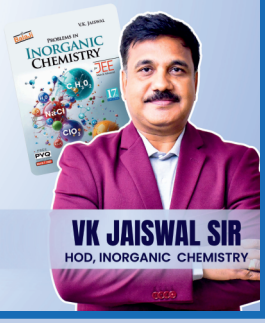


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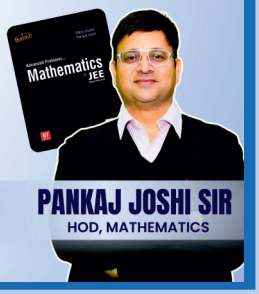


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MATHEMATICS

TARGET IIT JEE
MEGA-XII

INDEFINITE
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DEFINITE INTEGRATION

CONTENTS

KEY CONCEPT	*Page –2*
EXERCISE–I	*Page –7*
EXERCISE–II	*Page –9*
EXERCISE–III	*Page –12*
EXERCISE–IV	*Page –14*
EXERCISE–V	*Page –18*
ANSWER KEY	*Page –24-27*

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KEY CONCEPTS

1. DEFINITION :

If f & g are functions of x such that g′(x) = f(x) then the function g is called a **PRIMITIVE OR ANTIDERIVATIVE OR INTEGRAL** of f(x) w.r.t. x and is written symbolically as

$\int f(x)\,dx = g(x) + c \Leftrightarrow \frac{d}{dx}\{g(x) + c\} = f(x)$, where c is called the **constant of integration**.

2. STANDARD RESULTS :

(i) $\int (ax + b)^n\,dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c \quad n \neq -1$

(ii) $\int \frac{dx}{ax+b} = \frac{1}{a}\,\ell n\,(ax + b) + c$

(iii) $\int e^{ax+b}\,dx = \frac{1}{a}\,e^{ax+b} + c$

(iv) $\int a^{px+q}\,dx = \frac{1}{p}\frac{a^{px+q}}{\ell na} \quad (a > 0) + c$

(v) $\int \sin(ax + b)\,dx = -\frac{1}{a}\cos(ax + b) + c$

(vi) $\int \cos(ax + b)\,dx = \frac{1}{a}\sin(ax + b) + c$

(vii) $\int \tan(ax + b)\,dx = \frac{1}{a}\,\ell n\sec(ax + b) + c$

(viii) $\int \cot(ax + b)\,dx = \frac{1}{a}\,\ell n\sin(ax + b) + c$

(ix) $\int \sec^2(ax + b)\,dx = \frac{1}{a}\tan(ax + b) + c$

(x) $\int \operatorname{cosec}^2(ax + b)\,dx = -\frac{1}{a}\cot(ax + b) + c$

(xi) $\int \sec(ax + b)\,.\,\tan(ax + b)\,dx = \frac{1}{a}\sec(ax + b) + c$

(xii) $\int \operatorname{cosec}(ax + b)\,.\,\cot(ax + b)\,dx = -\frac{1}{a}\operatorname{cosec}(ax + b) + c$

(xiii) $\int \sec x\,dx = \ln(\sec x + \tan x) + c$ **OR** $\ln\tan\left(\frac{\pi}{4} + \frac{x}{2}\right) + c$

(xiv) $\int \operatorname{cosec} x\,dx = \ln(\operatorname{cosec} x - \cot x) + c$ **OR** $\ln\tan\frac{x}{2} + c$ **OR** $-\ln(\operatorname{cosec} x + \cot x)$

(xv) $\int \sinh x\,dx = \cosh x + c$

(xvi) $\int \cosh x\,dx = \sinh x + c$

(xvii) $\int \operatorname{sech}^2 x\,dx = \tanh x + c$

(xviii) $\int \operatorname{cosech}^2 x\,dx = -\coth x + c$

(xix) $\int \operatorname{sech} x\,.\,\tanh x\,dx = -\operatorname{sech} x + c$

(xx) $\int \operatorname{cosech} x\,.\,\coth x\,dx = -\operatorname{cosech} x + c$

(xxi) $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + c$

(xxii) $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + c$

(xxiii) $\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\frac{x}{a} + c$

(xxiv) $\int \frac{dx}{\sqrt{x^2+a^2}} = \ln\left[x+\sqrt{x^2+a^2}\right]$ **OR** $\sinh^{-1}\frac{x}{a} + c$

(xxv) $\int \frac{dx}{\sqrt{x^2-a^2}} = \ln\left[x+\sqrt{x^2-a^2}\right]$ **OR** $\cosh^{-1}\frac{x}{a} + c$

(xxvi) $\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\ln\frac{a + x}{a - x} + c$

(xxvii) $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\frac{x - a}{x + a} + c$

(xxviii) $\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + c$

(xxix) $\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\ln\left[x+\sqrt{x^2+a^2}\right] + c$

(xxx) $\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ln\left[x+\sqrt{x^2-a^2}\right] + c$

(xxxi) $\int e^{ax}.\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx) + c$

(xxxii) $\int e^{ax}.\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx) + c$

3. TECHNIQUES OF INTEGRATION :

(i) **Substitution** or change of independent variable.

Integral $I = \int f(x)\,dx$ is changed to $\int f(\phi(t))\,f'(t)\,dt$, by a suitable substitution $x = \phi(t)$ provided the later integral is easier to integrate .

(ii) **Integration by part :** $\int u.v\,dx = u\int v\,dx - \int\left[\frac{du}{dx}.\int v dx\right]dx$ where u & v are differentiable function .

Note : While using integration by parts, choose u & v such that

(a) $\int v\,dx$ is simple & **(b)** $\int\left[\frac{du}{dx}.\int v dx\right]dx$ is simple to integrate.

This is generally obtained, by keeping the order of u & v as per the order of the letters in **ILATE**, where ;
I – Inverse function, L – Logarithmic function ,
A – Algebraic function, T – Trigonometric function & E – Exponential function

(iii) **Partial fraction** , spiliting a bigger fraction into smaller fraction by known methods .

4. INTEGRALS OF THE TYPE :

(i) $\int [f(x)]^n f'(x)\,dx$ **OR** $\int \frac{f'(x)}{[f(x)]^n}\,dx$ put $f(x) = t$ & proceed .

(ii) $\int \frac{dx}{ax^2+bx+c}$, $\int \frac{dx}{\sqrt{ax^2+bx+c}}$, $\int \sqrt{ax^2+bx+c}\,dx$

Express $ax^2 + bx + c$ in the form of perfect square & then apply the standard results .

(iii) $\int \frac{px+q}{ax^2+bx+c}\,dx$, $\int \frac{px+q}{\sqrt{ax^2+bx+c}}\,dx$.

Express $px + q = A$ (differential co-efficient of denominator) + B .

(iv) $\int e^x[f(x) + f'(x)]\,dx = e^x.f(x) + c$ **(v)** $\int [f(x) + xf'(x)]\,dx = x\,f(x) + c$

(vi) $\int \frac{dx}{x(x^n+1)}$ $n \in N$ Take x^n common & put $1 + x^{-n} = t$.

(vii) $\int \frac{dx}{x^2(x^n+1)^{(n-1)/n}}$ $n \in N$, take x^n common & put $1+x^{-n} = t^n$

(viii) $\int \frac{dx}{x^n(1+x^n)^{1/n}}$ take x^n common as x and put $1 + x^{-n} = t$.

(ix) $\int \frac{dx}{a+b\sin^2 x}$ **OR** $\int \frac{dx}{a+b\cos^2 x}$ **OR** $\int \frac{dx}{a\sin^2 x + b\sin x\cos x + c\cos^2 x}$

Multiply N^{r} & D^{r} by $\sec^2 x$ & put $\tan x = t$.

(x) $\int \frac{dx}{a+b\sin x}$ **OR** $\int \frac{dx}{a+b\cos x}$ **OR** $\int \frac{dx}{a + b\sin x + c\cos x}$

Hint : Convert sines & cosines into their respective tangents of half the angles , put $\tan \frac{x}{2} = t$

(xi) $\int \frac{a.\cos x + b.\sin x + c}{\ell.\cos x + m.\sin x + n}\, dx$. Express $Nr \equiv A(Dr) + B \frac{d}{dx}(Dr) + c$ & proceed .

(xii) $\int \frac{x^2+1}{x^4+Kx^2+1}\, dx$ **OR** $\int \frac{x^2-1}{x^4+Kx^2+1}\, dx$ where K is any constant .

Hint : Divide Nr & Dr by x^2 & proceed .

(xiii) $\int \frac{dx}{(ax+b)\sqrt{px+q}}$ & $\int \frac{dx}{(ax^2+bx+c)\sqrt{px+q}}$; put $px + q = t^2$.

(xiv) $\int \frac{dx}{(ax+b)\sqrt{px^2+qx+r}}$, put $ax + b = \frac{1}{t}$; $\int \frac{dx}{(ax^2+bx+c)\sqrt{px^2+qx+r}}$, put $x = \frac{1}{t}$

(xv) $\int \sqrt{\frac{x-\alpha}{\beta-x}}\, dx$ or $\int \sqrt{(x-\alpha)(\beta-x)}$; put $x = \alpha\cos^2\theta + \beta\sin^2\theta$

$\int \sqrt{\frac{x-\alpha}{x-\beta}}\, dx$ or $\int \sqrt{(x-\alpha)(x-\beta)}$; put $x = \alpha\sec^2\theta - \beta\tan^2\theta$

$\int \frac{dx}{\sqrt{(x-\alpha)(x-\beta)}}$; put $x - \alpha = t^2$ or $x - \beta = t^2$.

DEFINITE INTEGRAL

1. $\int_a^b f(x)\,dx = F(b) - F(a)$ where $\int f(x)\,dx = F(x) + c$

VERY IMPORTANT NOTE : If $\int_a^b f(x)\,dx = 0 \Rightarrow$ then the equation $f(x) = 0$ has atleast one root lying in (a, b) provided f is a continuous function in (a, b) .

2. **PROPERTIES OF DEFINITE INTEGRAL :**

P–1 $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$ provided f is same **P – 2** $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

P–3 $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$, where c may lie inside or outside the interval $[a, b]$. This property to be used when f is piecewise continuous in (a, b) .

P–4 $\int_{-a}^{a} f(x)\,dx = 0$ if $f(x)$ is an odd function i.e. $f(x) = -f(-x)$.

$= 2\int_0^a f(x)\,dx$ if $f(x)$ is an even function i.e. $f(x) = f(-x)$.

P–5 $\int_a^b f(x)\,dx = \int_a^b f(a + b - x)\,dx$, In particular $\int_0^a f(x)\,dx = \int_0^a f(a - x)dx$

P–6 $\int_0^{2a} f(x)\,dx = \int_0^a f(x)\,dx + \int_0^a f(2a - x)\,dx = 2\int_0^a f(x)\,dx$ if $f(2a - x) = f(x)$

$= 0$ if $f(2a - x) = -f(x)$

P–7 $\int_0^{na} f(x)\,dx = n\int_0^a f(x)\,dx$; where'a'is the period of the function i.e. $f(a + x) = f(x)$

P–8 $\int_{a+nT}^{b+nT} f(x)\,dx = \int_a^b f(x)\,dx$ where $f(x)$ is periodic with period T & $n \in I$.

P–9 $\int_{ma}^{na} f(x)\,dx = (n - m)\int_0^a f(x)\,dx$ if $f(x)$ is periodic with period 'a' .

P–10 If $f(x) \le \phi(x)$ for $a \le x \le b$ then $\int_a^b f(x)\,dx \le \int_a^b \phi(x)\,dx$

P–11 $\left|\int_a^b f(x)dx\right| \le \int_a^b |f(x)|\,dx$.

P–12 If $f(x) \ge 0$ on the interval $[a, b]$, then $\int_a^b f(x)\,dx \ge 0$.

3. WALLI'S FORMULA :

$$\int_0^{\pi/2} \sin^n x \cdot \cos^m x \, dx = \frac{[(n-1)(n-3)(n-5)....1or2][(m-1)(m-3)....1or2]}{(m+n)(m+n-2)(m+n-4)....1or2} K$$

Where K = $\frac{\pi}{2}$ if both m and n are even (m, n $\in$ N) ;

= 1 otherwise

4. DERIVATIVE OF ANTIDERIVATIVE FUNCTION :

If h(x) & g(x) are differentiable functions of x then ,

$$\frac{d}{dx}\int_{g(x)}^{h(x)} f(t)\,dt = f[h(x)] \cdot h'(x) - f[g(x)] \cdot g'(x)$$

5. DEFINITE INTEGRAL AS LIMIT OF A SUM :

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} h\,[f(a) + f(a+h) + f(a+2h) + ... + f\left(a + \overline{n-1}\,h\right)]$$

$$= \lim_{h\to 0} h \sum_{r=0}^{n-1} f(a+rh) \quad \text{where } b - a = nh$$

If a = 0 & b = 1 then , $\lim_{n\to\infty} h \sum_{r=0}^{n-1} f(rh) = \int_0^1 f(x)\,dx$; where nh = 1 **OR**

$$\lim_{n\to\infty} \left(\frac{1}{n}\right) \sum_{r=1}^{n-1} f\left(\frac{r}{n}\right) = \int_0^1 f(x)\,dx \quad .$$

6. ESTIMATION OF DEFINITE INTEGRAL :

(i) For a monotonic decreasing function in (a , b) ; $f(b).(b-a) < \int_a^b f(x)\,dx < f(a).(b-a)$ &

(ii) For a monotonic increasing function in (a , b) ; $f(a).(b-a) < \int_a^b f(x)\,dx < f(b).(b-a)$

(iii) **Cauchy Scwarz Inequality :** If f and g are continuous functions on [a, b], then

$$\left(\int_a^b (f(t)g(t))dt\right)^2 \le \left(\int_a^b (f(t))^2 dt\right)\left(\int_a^b (g(t))^2 dt\right)$$

7. SOME IMPORTANT EXPANSIONS :

(i) $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} + ... \infty = \ln 2$

(ii) $\frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + ... \infty = \frac{\pi^2}{6}$

(iii) $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + ... \infty = \frac{\pi^2}{12}$

(iv) $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + ... \infty = \frac{\pi^2}{8}$

(v) $\frac{1}{2^2} + \frac{1}{4^2} + \frac{1}{6^2} + \frac{1}{8^2} + ... \infty = \frac{\pi^2}{24}$

EXERCISE–I

1. $\int \frac{\tan 2\theta}{\sqrt{\cos^6\theta+\sin^6\theta}}d\theta$ **2.** $\int \frac{5x^4+4x^5}{\left(x^5+x+1\right)^2}dx$ **3.** $\int \frac{\cos^2 x}{1+\tan x}dx$

4. $\int\left(\frac{\sin x+\sin 3x+\sin 5x+\sin 7x+\sin 9x+\sin 11x+\sin 13x+\sin 15x}{\cos x+\cos 3x+\cos 5x+\cos 7x+\cos 9x+\cos 11x+\cos 13x+\cos 15x}\right)dx$

5. $\int \frac{\ell n\left(\ell n\left(\frac{1+x}{1-x}\right)\right)}{1-x^2}dx$ **6.** $\int\left[\left(\frac{x}{e}\right)^x+\left(\frac{e}{x}\right)^x\right]\ell n x\,dx$

7. $\int \cos 2\theta \,.\, \ell n\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}d\theta$ **8.** $\int \frac{a^2\sin^2 x+b^2\cos^2 x}{a^4\sin^2 x+b^4\cos^2 x}dx$

9. $\int \frac{dx}{\left(x+\sqrt{x(1+x)}\right)^2}$ **10.** $\int \frac{x^3+x+1}{x^4+x^2+1}dx$

11. $\int \sqrt{\frac{\sin(x-a)}{\sin(x+a)}}dx$ **12.** $\int (\sin x)^{-11/3}(\cos x)^{-1/3}dx$ **13.** $\int \frac{\cot x\,dx}{(1-\sin x)(\sec x+1)}$

14. $\int \sin^{-1}\sqrt{\frac{x}{a+x}}\,dx$ **15.** $\int\left[\frac{\sqrt{x^2+1}\left[\ln\left(x^2+1\right)-2\ln x\right]}{x^4}\right]dx$ **16.** $\int \frac{x+1}{x\left(1+xe^x\right)^2}dx$

17. Let $f(x)$ is a quadratic function such that $f(0) = 1$ and $\int \frac{f(x)\,dx}{x^2(x+1)^3}$ is a rational function, find the value of $f'(0)$.

18. Integrate $\frac{1}{2}$ f′(x) w.r.t. x^4, where $f(x) = \tan^{-1}x + \ell n\sqrt{1+x} - \ell n\sqrt{1-x}$

19. $\int \frac{(\sqrt{x}+1)dx}{\sqrt{x}\,(\sqrt[3]{x}+1)}$ **20.** $\int \frac{dx}{\sin\frac{x}{2}\sqrt{\cos^3\frac{x}{2}}}$ **21.** $\int \frac{x^2+x}{(e^x+x+1)^2}dx$

22. $\int \sqrt{\frac{\operatorname{cosec}x-\cot x}{\operatorname{cosec}x+\cot x}}\cdot\frac{\sec x}{\sqrt{1+2\sec x}}\,dx$ **23.** $\int \frac{\cos x-\sin x}{7-9\sin 2x}\,dx$ **24.** $\int \frac{dx}{\sec x+\operatorname{cosec}x}\,dx$

25. $\int \frac{dx}{\sin x+\sec x}$ **26.** $\int \tan x.\tan 2x.\tan 3x\,dx$ **27.** $\int \frac{dx}{\sqrt{\sin^3 x\sin(x+\alpha)}}$

28. $\int \frac{x^2}{(x\cos x-\sin x)(x\sin x+\cos x)}dx$ **29.** $\int \frac{3+4\sin x+2\cos x}{3+2\sin x+\cos x}\,dx$

30. $\int \frac{x^5+3x^4-x^3+8x^2-x+8}{x^2+1}dx$ **31.** $\int \frac{\sqrt{\sin^4 x+\cos^4 x}}{\sin^3 x\cos x}dx$, $x\in\left(0,\frac{\pi}{2}\right)$

32. $\int \frac{3x^2+1}{(x^2-1)^3}dx$ **33.** $\int \frac{e^{\cos x}(x\sin^3 x+\cos x)}{\sin^2 x}dx$ **34.** $\int \frac{(ax^2-b)\,dx}{x\sqrt{c^2x^2-(ax^2+b)^2}}$

35. $\int \frac{e^x\left(2-x^2\right)}{(1-x)\sqrt{1-x^2}}dx$ **36.** $\int \frac{x}{(7x-10-x^2)^{3/2}}\,dx$ **37.** $\int \frac{x\ln x}{\left(x^2-1\right)^{3/2}}\,dx$

38. $\int \sqrt{\frac{(1-\sin x)(2-\sin x)}{(1+\sin x)(2+\sin x)}}\,dx$

39. $\int \frac{\sqrt{\cot x}-\sqrt{\tan x}}{1+3\sin 2x}\,dx$

40. $\int \frac{4x^5-7x^4+8x^3-2x^2+4x-7}{x^2(x^2+1)^2}dx$

41. $\int \frac{dx}{(x-\alpha)\sqrt{(x-\alpha)(x-\beta)}}$

42. $\int \frac{dx}{\cos^3 x-\sin^3 x}$

43. $\int \frac{\sqrt{\cos 2x}}{\sin x}\,dx$

44. $\int \frac{(1+x^2)dx}{1-2x^2\cos\alpha+x^4}$ $\alpha \in (0, \pi)$

45. Evaluate the integral $\int \frac{\cos^2 x+\sin 2x}{(2\cos x-\sin x)^2}\,dx$ where $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

46. Evaluate the integral $\int \frac{(x^2-x^3)\,dx}{(x+1)\,(x^3+x^2+x)^{3/2}}$

47. Evaluate $\int \frac{\sin^3 x\,dx}{(\cos^4 x+3\cos^2 x+1)\tan^{-1}(\sec x+\cos x)}$

48. Evaluate $\int \frac{(x\cos x+1)\,dx}{\sqrt{2x^3e^{\sin x}+x^2}}$

49. $\int \frac{\sqrt[3]{(1+\sqrt[4]{x})}}{\sqrt{x}}\,dx$

Match the Column:

50.

	Column-I		**Column-II**
(A)	$\int \frac{x^4-1}{x^2\sqrt{x^4+x^2+1}}dx$	(P)	$\ln\left(\frac{(x^2+1)+\sqrt{x^4+1}}{x}\right)+C$
(B)	$\int \frac{x^2-1}{x\sqrt{1+x^4}}dx$	(Q)	$C-\frac{1}{\sqrt{2}}\ln\left(\frac{\sqrt{x^4+1}-\sqrt{2}x}{(x^2-1)}\right)$
(C)	$\int \frac{1+x^2}{(1-x^2)\sqrt{1+x^4}}dx$	(R)	$C-\tan^{-1}\left(\sqrt{\sqrt{1+\frac{1}{x^4}}-1}\right)$
(D)	$\int \frac{1}{(1+x^4)\sqrt{\sqrt{1+x^4}-x^2}}dx$	(S)	$\frac{\sqrt{x^4+x^2+1}}{x}+C$

EXERCISE–II

1. Evaluate: $\int_0^1 e^{\ln \tan^{-1} x} \cdot \sin^{-1}(\cos x)\, dx$.

2. Prove that

(a) $\int_\alpha^\beta \sqrt{(x-\alpha)(\beta-x)}\ dx = \frac{(\beta-\alpha)^2\pi}{8}$ **(b)** $\int_\alpha^\beta \sqrt{\frac{x-\alpha}{\beta-x}}\ dx = (\beta-\alpha)\frac{\pi}{2}$

(c) $\int_\alpha^\beta \frac{dx}{x\sqrt{(x-\alpha)(\beta-x)}} = \frac{\pi}{\sqrt{\alpha\beta}}$ where $\alpha, \beta > 0$ **(d)** $\int_\alpha^\beta \frac{x\,dx}{\sqrt{(x-\alpha)(\beta-x)}} = (\alpha+\beta)\frac{\pi}{2}$ where $\alpha < \beta$

3. (a) Evaluate $I_n = \int_1^e (\ln^n x)dx$ hence find I_3.

(b) Determine a positive integer $n \le 5$, such that $\int_0^1 e^x (x-1)^n\ dx = 16 - 6e$.

4. $\int_0^{\pi/2} \sin 2x \cdot \arctan(\sin x)\, dx$

5. If $P = \int_0^\infty \frac{x^2}{1+x^4}dx$; $Q = \int_0^\infty \frac{x\,dx}{1+x^4}$ and $R = \int_0^\infty \frac{dx}{1+x^4}$ then prove that

(a) $Q = \frac{\pi}{4}$, **(b)** $P = R$, **(c)** $P - \sqrt{2}\,Q + R = \frac{\pi}{2\sqrt{2}}$

6. $\int_1^2 \frac{(x^2-1)\,dx}{x^3 \cdot \sqrt{2x^4-2x^2+1}} = \frac{u}{v}$ where u and v are in their lowest form. Find the value of $\frac{(1000)u}{v}$.

7. Evaluate $\int_0^{\pi/2} \frac{\sin^6 x\, dx}{\sin x + \cos x}$

8. For $a \ge 2$, if the value of the definite integral $\int_0^\infty \frac{dx}{a^2 + (x - (1/x))^2}$ equals $\frac{\pi}{5050}$. Find the value of a.

9. If a_1, a_2 and a_3 are the three values of *a* which satisfy the equation

$$\int_0^{\pi/2} (\sin x + a\cos x)^3 dx - \frac{4a}{\pi-2}\int_0^{\pi/2} x\cos x\, dx = 2$$

then find the value of $1000(a_1^2 + a_2^2 + a_3^2)$.

10. Let $u = \int_0^{\pi/4} \left(\frac{\cos x}{\sin x + \cos x}\right)^2 dx$ and $v = \int_0^{\pi/4} \left(\frac{\sin x + \cos x}{\cos x}\right)^2 dx$. Find the value of $\frac{v}{u}$.

11. $\int_0^{\pi/2} \sqrt{\frac{1-\sin 2x}{1+\sin 2x}}dx$

12. $\int_{-2}^{2} \frac{x^2 - x}{\sqrt{x^2+4}}dx$

13. $\int_{-\sqrt{2}}^{\sqrt{2}} \frac{2x^7+3x^6-10x^5-7x^3-12x^2+x+1}{x^2+2}dx$

14. $\int_0^{\pi/4} \frac{x\,dx}{\cos x\,(\cos x + \sin x)}$

15. $\int_0^1 \frac{\sin^{-1}\sqrt{x}}{x^2-x+1}dx$

16. $\int_1^{\frac{1+\sqrt{5}}{2}} \frac{x^2+1}{x^4-x^2+1} ln\left(1+x-\frac{1}{x}\right)dx$

17. $\lim_{n\to\infty} n^2 \int_{-1/n}^{1/n} (2007\sin x + 2008\cos x)|x|dx$.

18. Find the value of the definite integral $\int_0^{\pi} \left|\sqrt{2}\sin x + 2\cos x\right| dx$.

19. If $\int_0^{\pi} \sqrt{(\cos x+\cos 2x+\cos 3x)^2+(\sin x+\sin 2x+\sin 3x)^2}\,dx$ has the value equal to $\left(\frac{\pi}{k}+\sqrt{w}\right)$ where k and w are positive integers find the value of $(k^2 + w^2)$.

20. $\int_0^1 \frac{1-x}{1+x}\cdot\frac{dx}{\sqrt{x+x^2+x^3}}$

21. $\int_0^{\pi/2} \frac{a\sin x+b\cos x}{\sin\left(\frac{\pi}{4}+x\right)}dx$

22. $\int_0^1 \frac{x^2.\ln x}{\sqrt{1-x^2}}dx$

23. If $\int_{\pi/4}^{\pi/3} \frac{(\sin^3\theta-\cos^3\theta-\cos^2\theta)(\sin\theta+\cos\theta+\cos^2\theta)^{2007}}{(\sin\theta)^{2009}(\cos\theta)^{2009}}d\theta = \frac{(a+\sqrt{b})^n-(1+\sqrt{c})^n}{d}$

where a, b, c and d are all positive integers. Find the value (a + b + c + d).

24. $\int_0^{\sqrt{3}} \sin^{-1}\frac{2x}{1+x^2}dx$

25. $\int_0^{\pi} \frac{(ax+b)\sec x\tan x}{4+\tan^2 x}dx$ (a,b>0)

26. $\int_0^{\pi} \frac{(2x+3)\sin x}{(1+\cos^2 x)}dx$

27. $\int_0^{\pi} \frac{dx}{(5+4\cos x)^2}$

28. $\int_1^{16} \tan^{-1}\sqrt{\sqrt{x}-1}\,dx$

29. $\int_0^{2\pi} \frac{dx}{2+\sin 2x}$

30. $\int_0^a \frac{ln(1+ax)}{1+x^2}dx$, $a\in N$

31. $\int_0^{\frac{ln 3}{2}} \frac{e^x+1}{e^{2x}+1}dx$

32. $\int_0^{2\pi} \frac{x^2\sin x}{8+\sin^2 x}dx$

33. Let α, β be the distinct positive roots of the equation tan x = 2x then evaluate $\int_0^1 (\sin\alpha x\cdot\sin\beta x)dx$, independent of α and β.

34. Show that $\int_0^{p+q\pi} |\cos x|\,dx = 2q + \sin p$ where $q \in N$ & $-\frac{\pi}{2} < p < \frac{\pi}{2}$

35. Show that the sum of the two integrals $\int_{-4}^{-5} e^{(x+5)^2}dx + 3\int_{1/3}^{2/3} e^{9(x-2/3)^2}dx$ is zero.

36. If $\int_0^{\pi} \frac{x\sin^3 x}{4-\cos^2 x}dx = \pi\left(1-\frac{a\,ln\,b}{c}\right)$ where a and b are prime and $c \in N$, find the value of (a + b + c).

37. $\int_0^{\pi/2} \tan^{-1}\left[\frac{\sqrt{1+\sin x}+\sqrt{1-\sin x}}{\sqrt{1+\sin x}-\sqrt{1-\sin x}}\right]dx$

38. $\int_{\frac{\sqrt{3a^2+b^2}}{2}}^{\frac{\sqrt{a^2+b^2}}{2}} \frac{x.dx}{\sqrt{(x^2-a^2)(b^2-x^2)}}$

39. Comment upon the nature of roots of the quadratic equation $x^2 + 2x = k + \int_0^1 |t + k|\,dt$ depending on the value of $k \in R$.

40. Evaluate the definite integral, $\int_{-1}^{1} \frac{(2x^{332} + x^{998} + 4x^{1668} \cdot \sin x^{691})}{1 + x^{666}} dx$

41. $\int_0^{\pi} \frac{x^2 \sin 2x . \sin\left(\frac{\pi}{2} . \cos x\right)}{2x - \pi} dx$

42. (a) Show that $\int_0^{\infty} \frac{dx}{x^2 + 2x\cos\theta + 1} = 2\int_0^{1} \frac{dx}{x^2 + 2x\cos\theta + 1}$

(b) Evaluate: $f(\theta) = \int_0^{\infty} \frac{\tan^{-1} x}{x^2 + 2x\cos\theta + 1} dx$, $\theta \in (0, \pi)$

43. Evaluate: $\lim_{n\to\infty} \frac{1}{n^2} \sum_{k=0}^{n-1} \left[k \int_k^{k+1} \sqrt{(x-k)(k+1-x)}\, dx \right]$

44. Show that $\int_0^{\infty} f\left(\frac{a}{x} + \frac{x}{a}\right) . \frac{\ln x}{x} dx = \ln a . \int_0^{\infty} f\left(\frac{a}{x} + \frac{x}{a}\right) . \frac{dx}{x}$

45. Let $y = f(x)$ be a quadratic function with $f'(2) = 1$. Find the value of the integral $\int_{2-\pi}^{2+\pi} f(x) \cdot \sin\left(\frac{x-2}{2}\right) dx$.

46. Prove that $\int_0^{\pi} \frac{\ln(1 - \lambda^2 \sin^2 x)}{\sin x} dx = -2(\sin^{-1}\lambda)^2$ if $0 \le \lambda \le 1$.

47. Evaluate $\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{\cos^{-1}\left(\frac{2x}{x^2+1}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right)}{e^x + 1} dx$

48. Prove that $\int_0^x \left(\int_0^u f(t)\, dt \right) du = \int_0^x f(u)(x - u)\, du$

49. Prove that $\int_0^{\pi/4} \frac{x^2}{(x\sin x + \cos x)^2} dx = \frac{4 - \pi}{4 + \pi}$

EXERCISE–III

1. If the derivative of f(x) wrt x is $\frac{\cos x}{f(x)}$ then show that f(x) is a periodic function.

2. Find the range of the function, $f(x) = \int_{-1}^{1} \frac{\sin x \; dt}{1 - 2t\cos x + t^2}$.

3. A function f is defined in [–1 , 1] as $f'(x) = 2x \sin\frac{1}{x} - \cos\frac{1}{x}$; $x \neq 0$; f(0) = 0; f (1/π) = 0. Discuss the continuity and derivability of f at x = 0.

4. Let $f(x) = \begin{cases} -1 & \text{if } -2 \le x \le 0 \\ |x-1| & \text{if } 0 < x \le 2 \end{cases}$ and $g(x) = \int_{-2}^{x} f(t)\,dt$. Define g (x) as a function of x and test the continuity and differentiability of g(x) in (–2, 2).

5. If $\phi(x) = \cos x - \int_0^x (x-t)\,\phi(t)\,dt$. Then find the value of $\phi''(x) + \phi(x)$.

6. If $y = \frac{1}{a}\int_0^x f(t)\cdot\sin a(x-t)\,dt$ then prove that $\frac{d^2y}{dx^2} + a^2y = f(x)$.

7. If $y = x^{\int_1^x \ln t\,dt}$, find $\frac{dy}{dx}$ at x = e.

8. A curve C_1 is defined by: $\frac{dy}{dx} = e^x \cos x$ for $x \in [0, 2\pi]$ and passes through the origin. Prove that the roots of the function y = 0 (other than zero) occurs in the ranges $\frac{\pi}{2} < x < \pi$ and $\frac{3\pi}{2} < x < 2\pi$.

9. (a) Let $g(x) = x^c \cdot e^{2x}$ & let $f(x) = \int_0^x e^{2t} \cdot (3t^2 + 1)^{1/2}\,dt$. For a certain value of 'c', the limit of $\frac{f'(x)}{g'(x)}$ as $x \to \infty$ is finite and non zero. Determine the value of 'c' and the limit.

(b) Find the constants 'a' (a > 0) and 'b' such that, $\lim_{x\to 0} \frac{\int_0^x \frac{t^2\,dt}{\sqrt{a+t}}}{bx - \sin x} = 1$.

10. Evaluate: $\lim_{x\to+\infty} \frac{d}{dx} \int_{2\sin\frac{1}{x}}^{3\sqrt{x}} \frac{3t^4+1}{(t-3)(t^2+3)}\,dt$

11. Determine a pair of number *a* and *b* for which $\int_0^1 \frac{ax+b}{(x^2+3x+2)^2}\,dx = \frac{5}{2}$.

12. If $\int_0^\infty \frac{\ell nt}{x^2+t^2}dt = \frac{\pi\,\ell n\,2}{4}$ (x > 0) then show that there can be two integral values of 'x' satisfying this equation.

13. Evaluate: $\lim_{a\to\infty} \frac{\int_0^a \sin^4 x\,dx}{a}$

14. Prove that : (a) $I_{m,n} = \int_0^1 x^m \cdot (1-x)^n\,dx = \frac{m!n!}{(m+n+1)!}$ m , n ∈ N.

(b) $I_{m,n} = \int_0^1 x^m \cdot (\ln x)^n\,dx = (-1)^n \frac{n!}{(m+1)^{n+1}}$ m , n ∈ N.

15. Find a positive real valued continuously differentiable functions *f* on the real line such that for all x

$$f^2(x) = \int_0^x \left((f(t))^2 + (f'(t))^2 \right) dt + e^2$$

16. Let f(x) be a continuously differentiable function then prove that, $\int_1^x [t]\, f'(t)\, dt = [x].\, f(x) - \sum_{k=1}^{[x]} f(k)$ where $[\cdot]$ denotes the greatest integer function and x > 1.

17. Let $F(x) = \int_{-1}^{x} \sqrt{4+t^2}\, dt$ and $G(x) = \int_{x}^{1} \sqrt{4+t^2}\, dt$ then compute the value of (FG)' (0) where dash denotes the derivative.

18. Show that for a continuously thrice differentiable function f(x)

$$f(x) - f(0) = xf'(0) + \frac{f''(0).x^2}{2} + \frac{1}{2}\int_0^x f'''(t)(x-t)^2\, dt$$

19. Evaluate: **(a)** $\lim_{n\to\infty} \left[\left(1+\frac{1}{n^2}\right)\left(1+\frac{2^2}{n^2}\right)\left(1+\frac{3^2}{n^2}\right)..... \left(1+\frac{n^2}{n^2}\right) \right]^{1/n}$; **(b)** $\lim_{n\to\infty} \frac{1}{n} \left[\frac{1}{n+1} + \frac{2}{n+2} + ... + \frac{3n}{4n} \right]$

20. **(a)** $\lim_{n\to\infty} \left[\frac{n!}{n^n} \right]^{1/n}$

(b) For positive integers *n*, let $A_n = \frac{1}{n}\{(n+1)+(n+2)+.....(n+n)\}$,

$B_n = \{(n+1)(n+2).....(n+n)\}^{1/n}$. If $\lim_{n\to\infty} \frac{A_n}{B_n} = \frac{ae}{b}$ where a, b ∈ N and relatively prime find the value of (a + b).

21. Let f be an injective function such that f(x) f(y) + 2 = f(x) + f(y) + f(xy) for all non negative real x & y with f′(0) = 0 & f′(1) = 2 ≠ f(0). Find f(x) & show that, $3\int f(x)\, dx - x(f(x) + 2)$ is a constant.

22. Let $I = \int_{1/2}^{2} \frac{\ln t}{1+t^n} dt$, find the sign of the integral for different values of $n \in N \cup \{0\}$.

23. Let f be a function such that $|f(u) - f(v)| \le |u - v|$ for all real u & v in an interval [a, b]. Then:

(i) Prove that f is continuous at each point of [a, b].

(ii) Assume that f is integrable on [a, b]. Prove that, $\left| \int_a^b f(x)\, dx - (b-a) f(c) \right| \le \frac{(b-a)^2}{2}$, where $a \le c \le b$

24. Prove that $\sum_{k=0}^{n} (-1)^k \binom{n}{k} \frac{1}{k+m+1} = \sum_{k=0}^{m} (-1)^k \binom{m}{k} \frac{1}{k+n+1}$

25. Let *f* and *g* be function that are differentiable for all real numbers x and that have the following properties:

(i) $f'(x) = f(x) - g(x)$; (ii) $g'(x) = g(x) - f(x)$

(iii) $f(0) = 5$; (iv) $g(0) = 1$

(a) Prove that $f(x) + g(x) = 6$ for all x.

(b) Find $f(x)$ and $g(x)$.

26. If $f(x) = x + \int_0^1 (xy^2 + x^2y) f(y)\, dy$ where x and y are independent variable. Find f(x).

27. Prove that $\sin x + \sin 3x + \sin 5x + ... + \sin (2k-1)x = \frac{\sin^2 kx}{\sin x}$, $k \in N$ and hence prove that, $\int_0^{\pi/2} \frac{\sin^2 kx}{\sin x}\, dx = 1+\frac{1}{3}+\frac{1}{5}+\frac{1}{7}+.....+\frac{1}{2k-1}$.

28. If $U_n = \int_0^{\pi/2} \frac{\sin^2 nx}{\sin^2 x}\, dx$, then show that $U_1, U_2, U_3, ..., U_n$ constitute an AP.
Hence or otherwise find the value of U_n.

29. Suppose $f : R \to R^+$ be a differentiable function and satisfies $3f(x+y) = f(x) \cdot f(y)$ for all $x, y \in R$ with $f(1) = 6$. If $U = \lim_{n\to\infty} n\left(f\left(1+\frac{1}{n}\right) - f(1)\right)$ and $V = \int_0^3 f(x)\,dx$ then find

(a) the range of f (x); **(b)** the value of U ; **(c)** the value of the product UV

30. Prove the inequalities:

(a) $\frac{\pi}{6} < \int_0^1 \frac{dx}{\sqrt{4-x^2-x^3}} < \frac{\pi\sqrt{2}}{8}$

(b) $2e^{-1/4} < \int_0^2 e^{x^2-x}\, dx < 2e^2$.

(c) $\frac{1}{3} < \int_0^1 x^{(\sin x+\cos x)^2} dx < \frac{1}{2}$

(d) $\frac{1}{2} \le \int_0^2 \frac{dx}{2+x^2} \le \frac{5}{6}$

EXERCISE–IV

1. $\int \frac{dx}{\cos x + \sqrt{3}\sin x}$ is equal to **(AIEEE 2007)**

(A) $\frac{1}{2}\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)+c$ (B) $\frac{1}{2}\log\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)+c$

(C) $\log\tan\left(\frac{x}{2}+\frac{\pi}{12}\right)+c$ (D) $\log\tan\left(\frac{x}{2}-\frac{\pi}{12}\right)+c$

2. Let $F(x) = f(x) + f\left(\frac{1}{x}\right)$, where $f(x) = \int_1^x \frac{\log t}{1+t} dt$. Then F (e) is equal to : **(AIEEE 2007)**

(A) $\frac{1}{2}$ (B) 0 (C) 1 (D) 2

3. The value of $\sqrt{2}\int \frac{\sin x\, dx}{\sin\left(x-\frac{\pi}{4}\right)}$ is : **(AIEEE 2008)**

(A) $x+\log\left|\cos\left(x-\frac{\pi}{4}\right)\right|+c$ (B) $x-\log\left|\sin\left(x-\frac{\pi}{4}\right)\right|+c$

(C) $x+\log\left|\sin\left(x-\frac{\pi}{4}\right)\right|+c$ (D) $x-\log\left|\cos\left(x-\frac{\pi}{4}\right)\right|+c$

4. $\int_0^{\pi} [\cot x]dx$, where [.] denotes the greatest integer function, is equal to : **(AIEEE 2009)**

(A) $\frac{\pi}{2}$ (B) 1 (C) –1 (D) $-\frac{\pi}{2}$

5. Let p(x) be a function defined on R such that p'(x) = p'(1 – x), for all $x \in [0, 1]$, p(0) = 1, and p(1) = 41. Then $\int_0^1 p(x)dx$ is equal to : **(AIEEE 2010)**

(A) 42 (B) $\sqrt{41}$ (C) 21 (D) 41

6. The value of $\int_0^1 \frac{8\log(1+x)}{1+x^2}dx$ is : **(AIEEE 2011)**

(A) log 2 (B) π log 2 (C) $\frac{\pi}{8}\log 2$ (D) $\frac{\pi}{2}\log 2$

7. For $x \in \left(0, \frac{5\pi}{2}\right)$, define $f(x) = \int_0^x \sqrt{t}\sin t\,dt$. Then f has : **(AIEEE 2011)**

(A) Local maximum at π and local minimum at 2π (B) Local maximum at π and 2π
(C) Local minimum at π and 2π (D) Local minimum at π and local maximum at 2π

8. If $g(x) = \int_0^x \cos^4 t\,dt$, then g(x + π) equals : **(AIEEE 2012)**

(A) $\frac{g(x)}{g(\pi)}$ (B) g(x) + g(π) (C) g(x) – g(π) (D) g(x) . g(π)

9. If $\int \frac{5\tan x}{\tan x - 2}dx = x + a\ln(\sin x - 2\cos x) + k$, where k is some constant, then a = **(AIEEE 2012)**

(A) –1 (B) –2 (C) 1 (D) 2

10. If $\int f(x)\,dx = \Psi(x)$, then $\int x^5 f(x^3)dx$ is equal to : **[JEE Main 2013]**

(A) $\frac{1}{3}\left[x^3\Psi(x^3) - \int x^3\Psi(x^3)dx\right] + C$ (B) $\frac{1}{3}\left[x^3\Psi(x^3) - \int x^2\Psi(x^3)dx\right] + C$

(C) $\frac{1}{3}x^3\Psi(x^3) - 3\int x^3\Psi(x^3)dx + C$ (D) $\frac{1}{3}x^3\Psi(x^3) - \int x^2\Psi(x^3)dx + C$

11. Statement-1 : The value of the integral $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1+\sqrt{\tan x}}$ is equal to $\frac{\pi}{6}$ **[JEE Main 2013]**

Statement-2 : $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$.

(A) Statement-1 is false ; Statement-2 is true.

(B) Statement-1 is true ; Statement-2 is true ; Statement-2 is a **correct** explanation for Statement-1

(C) Statement-1 is true ; Statement-2 is true ; Statement-2 is **not** a **correct** explanation for Statement-1

(D) Statement-1 is true ; Statement-2 is false

12. The integral $\int\left(1+x-\frac{1}{x}\right)e^{x+\frac{1}{x}}dx$ is equal to **[JEE Main 2014]**

(A) $-x\,e^{x+\frac{1}{x}}+c$ (B) $(x-1)\,e^{x+\frac{1}{x}}+c$ (C) $x\,e^{x+\frac{1}{x}}+c$ (D) $(x+1)\,e^{x+\frac{1}{x}}+c$

13. The integral $\int_0^{\pi}\sqrt{1+4\sin^2\frac{x}{2}-4\sin\frac{x}{2}}\,dx$ equals : **[JEE Main 2014]**

(A) $4\sqrt{3}-4-\frac{\pi}{3}$ (B) $\pi-4$ (C) $\frac{2\pi}{3}-4-4\sqrt{3}$ (D) $4\sqrt{3}-4$

14. The integral $\int\frac{dx}{x^2(x^4+1)^{3/4}}$ equals: **[JEE Main 2015]**

(A) $-\left(\frac{x^4+1}{x^4}\right)^{1/4}+c$ (B) $\left(\frac{x^4+1}{x^4}\right)^{1/4}+c$ (C) $(x^4+1)^{1/4}+c$ (D) $-(x^4+1)^{1/4}+c$

15. The integral $\int_2^4\frac{\log x^2}{\log x^2+\log(36-12x+x^2)}\,dx$ is equal to : **[JEE Main 2015]**

(A) 6 (B) 2 (C) 4 (D) 1

16. The integral $\int\frac{2x^{12}+5x^9}{(x^5+x^3+1)^3}\,dx$ is equal to : **[JEE Main 2016]**

(A) $\frac{-x^5}{(x^5+x^3+1)^2}+C$ (B) $\frac{x^{10}}{2(x^5+x^3+1)^2}+C$ (C) $\frac{x^5}{2(x^5+x^3+1)^2}+C$ (D) $\frac{-x^{10}}{2(x^5+x^3+1)^2}+C$

where C is an arbitrary constant

17. $\lim_{n\to\infty}\left(\frac{(n+1)(n+2).....3n}{n^{2n}}\right)^{1/n}$ is equal to : **[JEE Main 2016]**

(A) $\frac{18}{e^4}$ (B) $\frac{27}{e^2}$ (C) $\frac{9}{e^2}$ (D) $3\log 3-2$

18. The integral $\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} \frac{dx}{1+\cos x}$ is equal to : **[JEE Main 2017]**

(A) 4 (B) –1 (C) – 2 (D) 2

19. Let $I_n = \int \tan^n x\,dx, (n > 1)$. $I_4 + I_6 = a\tan^5 x + bx^5 + C$, where C is a constant of integration, then the ordered pair (a, b) is equal to : **[JEE Main 2017]**

(A) $\left(\frac{1}{5}, -1\right)$ (B) $\left(-\frac{1}{5}, 0\right)$ (C) $\left(-\frac{1}{5}, 1\right)$ (D) $\left(\frac{1}{5}, 0\right)$

20. The area (in sq. units) of the region $\{(x, y) : x \geq 0, x + y \leq 3, x^2 \leq 4y \text{ and } y \leq 1 + \sqrt{x}\}$ is :

(A) $\frac{7}{3}$ (B) $\frac{5}{2}$ (C) $\frac{59}{12}$ (D) $\frac{3}{2}$ **[JEE Main 2017]**

21. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^2 x}{1+2^x} dx$ is : **[JEE Main 2018]**

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{8}$ (C) $\frac{\pi}{2}$ (D) 4π

22. The integral $\int \frac{\sin^2 x \cos^2 x}{(\sin^5 x + \cos^3 x \sin^2 x + \sin^3 x \cos^2 x + \cos^5 x)^2} dx$ is equal to : **[JEE Main 2018]**

(A) $\frac{-1}{1+\cot^3 x} + C$ (B) $\frac{1}{3(1+\tan^3 x)} + C$ (C) $\frac{-1}{3(1+\tan^3 x)} + C$ (D) $\frac{1}{1+\cot^3 x} + C$

(where C is a constant of integration)

EXERCISE–V

1. The option(s) with the values of a and L that satisfy the following equation is(are)

$$\frac{\int_0^{4\pi} e^t(\sin^6 at + \cos^4 at)\,dt}{\int_0^{\pi} e^t(\sin^6 at + \cos^4 at)\,dt} = L \ ?$$ **[JEE Advanced 2015]**

(A) $a = 2, L = \frac{e^{4\pi}-1}{e^{\pi}-1}$ (B) $a = 2, L = \frac{e^{4\pi}+1}{e^{\pi}+1}$ (C) $a = 4, L = \frac{e^{4\pi}-1}{e^{\pi}-1}$ (D) $a = 4, L = \frac{e^{4\pi}+1}{e^{\pi}+1}$

2. Let $f(x) = 7\tan^8 x + 7\tan^6 x - 3\tan^4 x - 3\tan^2 x$ for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the correct expression(s) is(are)

(A) $\int_0^{\pi/4} xf(x)dx = \frac{1}{12}$ (B) $\int_0^{\pi/4} f(x)dx = 0$ **[JEE Advanced 2015]**

(C) $\int_0^{\pi/4} xf(x)dx = \frac{1}{6}$ (D) $\int_0^{\pi/4} f(x)dx = 1$

3. Let $f'(x) = \frac{192x^3}{2+\sin^4 \pi x}$ for all $x \in \mathbb{R}$ with $f\left(\frac{1}{2}\right) = 0$. If $m \le \int_{1/2}^{1} f(x)dx \le M$, then the possible values of m and M are **[JEE Advanced 2015]**

(A) m = 13, M = 24 (B) $m = \frac{1}{4}, M = \frac{1}{2}$ (C) m = –11, M = 0 (D) m = 1, M = 12

Paragraph for question no. 4 to 5 **[JEE Advanced 2015]**

Let $F : \mathbb{R} \to \mathbb{R}$ be a thrice differentiable function. Suppose that F(1) = 0, F(3) = –4 and $F'(x) < 0$ for all $x \in (1/2, 3)$. Let $f(x) = xF(x)$ for all $x \in \mathbb{R}$.

4. The correct statement(s) is(are)

(A) $f'(1) < 0$ (B) $f(2) < 0$

(C) $f'(x) \neq 0$ for any $x \in (1, 3)$ (D) $f'(x) = 0$ for some $x \in (1, 3)$

5. If $\int_1^3 x^2F'(x)dx = -12$ and $\int_1^3 x^3F''(x)dx = 40$, then the correct expression(s) is(are)

(A) $9f'(3) + f'(1) - 32 = 0$ (B) $\int_1^3 f(x)dx = 12$ (C) $9f'(3) - f'(1) + 32 = 0$ (D) $\int_1^3 f(x)dx = -12$

6. If $\alpha = \int_0^1 (e^{9x+3\tan^{-1}x})\left(\frac{12+9x^2}{1+x^2}\right)dx$, **[JEE Advanced 2015]**

where $\tan^{-1} x$ takes only principal values, then the value of $\left(\log_e |1+\alpha| - \frac{3\pi}{4}\right)$ is

7. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous odd function, which vanishes exactly at one point and $f(1) = \frac{1}{2}$. Suppose that

$F(x) = \int_{-1}^{x} f(t)\,dt$ for all $x \in [-1, 2]$ and $G(x) = \int_{-1}^{x} t\,|f(f(t))|\,dt$ for all $x \in [-1, 2]$. If $\lim_{x\to 1}\frac{F(x)}{G(x)} = \frac{1}{14}$, then the value

of $f\left(\frac{1}{2}\right)$ is **[JEE Advanced 2015]**

8. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2\cos x}{1+e^x}dx$ is equal to : **[JEE Advanced 2016]**

(A) $\frac{\pi^2}{4} - 2$ (B) $\frac{\pi^2}{4} + 2$ (C) $\pi^2 - e^{\frac{\pi}{2}}$ (D) $\pi^2 + e^{\frac{\pi}{2}}$

9. Let $f(x) = \lim_{n\to\infty}\left(\frac{n^n(x+n)\left(x+\frac{n}{2}\right).....\left(x+\frac{n}{n}\right)}{n!(x^2+n^2)\left(x^2+\frac{n^2}{4}\right).....\left(x^2+\frac{n^2}{n^2}\right)}\right)^{\frac{x}{n}}$, for all $x > 0$. Then **[JEE Advanced 2016]**

(A) $f\left(\frac{1}{2}\right) \geq f(1)$ (B) $f\left(\frac{1}{3}\right) \leq f\left(\frac{2}{3}\right)$ (C) $f'(2) \leq 0$ (D) $\frac{f'(3)}{f(3)} \geq \frac{f'(2)}{f(2)}$

10. The total number of distinct $x \in [0, 1]$ for which $\int_0^x \frac{t^2}{1+t^4}dt = 2x - 1$ is : **[JEE Advanced 2016]**

11. If $I = \sum_{k=1}^{98}\int_k^{k+1}\frac{k+1}{x(x+1)}dx$, then **[JEE Advanced 2017]**

(A) $I < \frac{49}{50}$ (B) $I < \log_e 99$ (C) $I > \frac{49}{50}$ (D) $I > \log_e 99$

12. If $g(x) = \int_{\sin x}^{\sin(2x)} \sin^{-1}(t)\,dt$, then **[JEE Advanced 2017]**

(A) $g'\left(\frac{\pi}{2}\right) = -2\pi$ (B) $g'\left(-\frac{\pi}{2}\right) = 2\pi$ (C) $g'\left(\frac{\pi}{2}\right) = 2\pi$ (D) $g'\left(-\frac{\pi}{2}\right) = -2\pi$

13. Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function such that $f(0) = 0$, $f\left(\frac{\pi}{2}\right) = 3$ and $f'(0) = 1$. **[JEE Advanced 2017]**

If $g(x) = \int_x^{\pi/2} [(\text{cosec}\, t) f'(t) - (\cot t)(\text{cosec}\, t) f(t)]\, dt$ for $x \in \left(0, \frac{\pi}{2}\right]$, then $\lim_{x \to 0} g(x) =$

14. Let $f : \mathbb{R} \to (0, 1)$ be a continuous funciton. Then, which of the following function(s) has(have) the value zero at some point in the interval (0, 1)? **[JEE Advanced 2017]**

(A) $x - \int_0^{\frac{\pi}{2}-x} f(t)\cos t\, dt$ (B) $x^9 - f(x)$ (C) $e^x - \int_0^x f(t)\sin t\, dt$ (D) $f(x) + \int_0^{\frac{\pi}{2}} f(t)\sin t\, dt$

15. For each positive integer n, let $y_n = \frac{1}{n}((n+1)(n+2).....(n+n))^{\frac{1}{n}}$. For $x \in \mathbb{R}$, let [x] be the greatest integer less than or equal to x. If $\lim_{x \to \infty} y_n = L$, then the value of [L] is _____. **[JEE Advanced 2018]**

16. The value of the integral $\int_0^{\frac{1}{2}} \frac{1+\sqrt{3}}{((x+1)^2(1-x)^6)^{\frac{1}{4}}}\, dx$ is _____. **[JEE Advanced 2018]**

17. If $I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(1+e^{\sin x})(2-\cos 2x)}$ then $27\, I^2$ equals_____ **[JEE Advanced 2019]**

18. For $a \in R$, $|a| > 1$, let $\lim_{n \to \infty} \left(\frac{1 + \sqrt[3]{2} + ... + \sqrt[3]{n}}{n^{7/3}\left(\frac{1}{(an+1)^2} + \frac{1}{(an+2)^2} + ... + \frac{1}{(an+n)^2} \right)} \right) = 54$

Then the possible value(s) of a is/are : **[JEE Advanced 2019]**
(A) – 9 (B) 8 (C) 7 (D) –6

19. The value of the integral $\int_0^{\pi/2} \frac{3\sqrt{\cos\theta}}{(\sqrt{\cos\theta} + \sqrt{\sin\theta})^5} d\theta$ equals___. **[JEE Advanced 2019]**

20. Which of the following inequalities is/are TRUE? **[JEE Advanced 2020]**

(A) $\int_0^1 x\cos x\, dx \geq \frac{3}{8}$ (B) $\int_0^1 x\sin x\, dx \geq \frac{3}{10}$ (C) $\int_0^1 x^2\cos x\, dx \geq \frac{1}{2}$ (D) $\int_0^1 x^2\sin x\, dx \geq \frac{2}{9}$

21. Let $f : \mathbf{R} \to \mathbf{R}$ be a differentiable function such that its derivative f′ is continuous and $f(\pi) = -6$.

If $F : [0, \pi] \to \mathbf{R}$ is defined by $F(x) = \int_0^x f(t)dt$, and if

$\int_0^{\pi} (f'(x) + F(x))\cos x\, dx = 2$, then the value of f(0) is _____. **[JEE Advanced 2020]**

22. Let $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to R$ be a continuous function such that

$f(0) = 1$ and $\int_0^{\frac{\pi}{3}} f(t)\,dt = 0$

Then which of the following statements is (are) **True** ? **[JEE Advanced 2021]**

(A) The equation $f(x) - 3\cos 3x = 0$ has at least one solution in $\left(0, \frac{\pi}{3}\right)$

(B) The equation $f(x) - 3\sin 3x = -\frac{6}{\pi}$ has at least one solution in $\left(0, \frac{\pi}{3}\right)$

(C) $\lim_{x\to 0} \frac{x\int_0^x f(t)\,dt}{1-e^{x^2}} = -1$

(D) $\lim_{x\to 0} \frac{\sin x\int_0^x f(t)\,dt}{x^2} = -1$

Question Stem for Question Nos. 23 to 24

Question Stem **[JEE Advanced 2021]**

Let $g_i : \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \to \mathbf{R}$, i = 1, 2, and $f : \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \to \mathbf{R}$ be functions such that

$g_1(x) = 1$, $g_2(x) = |4x - \pi|$ and $f(x) = \sin^2 x$, for all $x \in \left[\frac{\pi}{8}, \frac{3\pi}{8}\right]$

Define

$$S_i = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} f(x) \cdot g_i(x)\,dx, \quad i = 1, 2$$

23. The value of $\frac{16S_1}{\pi}$ is ..... .

24. The value of $\frac{48S_2}{\pi^2}$ is ..... .

25. For any real number x, let [x] denote the largest integer less than or equal to x. If $I = \int_0^{10} \left[\sqrt{\frac{10x}{x+1}}\right] dx$, then the value of 9I is ___. **[JEE Advanced 2021]**

26. Consider the equation $\int_1^e \frac{(\log_e x)^{1/2}}{x\left(a - (\log_e x)^{3/2}\right)^2} dx = 1$, $a \in (-\infty, 0) \cup (1, \infty)$.

Which of the following statements is/are TRUE? **[JEE Advanced 2022]**

(A) No a satisfies the above equation

(B) An integer a satisfies the above equation

(C) An irrational number a satisfies the above equation

(D) More than one a saisfy the above equation

27. The greatest integer less than or equal to $\int_1^2 \log_2(x^3+1)\,dx + \int_1^{\log_2 9} (2^x-1)^{\frac{1}{3}}\,dx$ is _____.

[JEE Advanced 2022]

28. For positive integer n, define $f(x) = n + \frac{16+5n-3n^2}{4n+3n^2} + \frac{32+n-3n^2}{8n+3n^2} + \frac{48-3n-3n^2}{12n+3n^2} + \ldots + \frac{25n-7n^2}{7n^2}$.

Then, the value of $\lim_{n\to\infty} f(x)$ is equal to : **[JEE Advanced 2022]**

(A) $3 + \frac{4}{3}\log_e 7$ (B) $4 - \frac{3}{4}\log_e\left(\frac{7}{3}\right)$ (C) $4 - \frac{4}{3}\log_e\left(\frac{7}{3}\right)$ (D) $3 + \frac{3}{4}\log_e 7$

29. Let $f : (0, 1) \to \mathbb{R}$ be the function defined as $f(x) = \sqrt{n}$ if $x \in \left[\frac{1}{n+1}, \frac{1}{n}\right)$ where $n \in \mathbb{N}$. Let $g : (0, 1) \to \mathbb{R}$ be a function such that $\int_{x^2}^{x} \sqrt{\frac{1-t}{t}}\,dt < g(x) < 2\sqrt{x}$ for all $x \in (0, 1)$. Then $\lim_{x\to 0} f(x)\,g(x)$ **[JEE Advanced 2023]**

(A) does NOT exist (B) is equal to 1 (C) is equal to 2 (D) is equal to 3

30. Let $f : [1, \infty) \to \mathbf{R}$ be a differentiable function such that $f(1) = \frac{1}{3}$ and $3\int_1^x f(t)\,dt = x\,f(x) - \frac{x^3}{3}$, $x \in [1, \infty)$. Let e denote the base of the natural logarithm Then the value of f(e) is : **[JEE Advanced 2023]**

(A) $\frac{e^2+4}{3}$ (B) $\frac{\log_e 4 + e}{3}$ (C) $\frac{4e^2}{3}$ (D) $\frac{e^2-4}{3}$

Paragraph (Q.31 to Q.32)

Let $f : \left[0, \frac{\pi}{2}\right] \to [0, 1]$ be the function defined by $f(x) = \sin^2 x$ and let $g : \left[0, \frac{\pi}{2}\right] \to [0, \infty)$ be the function defined by $g(x) = \sqrt{\frac{\pi x}{2} - x^2}$. **[JEE Advanced 2024]**

31. The value of $2\int_0^{\pi/2} f(x)g(x)dx - \int_0^{\pi/2} g(x)dx$ is _____.

32. The value of $\frac{16}{\pi^3}\int_0^{\pi/2} f(x)g(x)dx$ is _____.

33. Let the function $f : [1, \infty) \to \mathbb{R}$ be defined by

$$f(t) = \begin{cases} (-1)^{n+1}2, & \text{if } t = 2n-1, n \in \mathbb{N}, \\ \frac{(2n+1-t)}{2} f(2n-1) + \frac{(t-(2n-1))}{2} f(2n+1), & \text{if } 2n-1 < t < 2n+1, n \in \mathbb{N}. \end{cases}$$

Define $g(x) = \int_1^x f(t)\,dt$, $x \in (1, \infty)$. Let α denote the number of solutions of the equation $g(x) = 0$ in the interval $(1, 8]$ and $\beta = \lim_{x \to 1+} \frac{g(x)}{x-1}$. The the value of $\alpha + \beta$ is equal to_____. **[JEE Advanced 2024]**

34. If $\alpha = \int_{\frac{1}{2}}^{2} \frac{\tan^{-1} x}{2x^2 - 3x + 2}\,dx$, then the value of $\sqrt{7}\tan\left(\frac{2\alpha\sqrt{7}}{\pi}\right)$ is_____.

(Here, the inverse trigonometric function $\tan^{-1} x$ assumes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.) **[JEE Advanced 2025]**

ANSWER KEY

EXERCISE–I

Q.1 $\ln\left(\frac{1+\sqrt{1+3\cos^2 2\theta}}{\cos 2\theta}\right)$ + C

Q.2 C – $\frac{x+1}{x^5+x+1}$ or C + $\frac{x^5}{x^5+x+1}$

Q.3 $\frac{1}{4}\ln(\cos x+\sin x)+\frac{x}{2}+\frac{1}{8}(\sin 2x+\cos 2x)+C$

Q.4 $\frac{1}{8}\ln(\sec 8x)+C$

Q.5 $\frac{1}{2}\left[\ln\left(\frac{1+x}{1-x}\right)\cdot\ln\left(\ln\frac{1+x}{1-x}\right)-\ln\left(\frac{1+x}{1-x}\right)\right]$ + C

Q.6 $\left(\frac{x}{e}\right)^x-\left(\frac{e}{x}\right)^x+C$

Q.7 $\frac{1}{2}(\sin 2\theta)\ln\left(\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right)-\frac{1}{2}\ln(\sec 2\theta)+C$

Q.8 $\frac{1}{a^2+b^2}\left(x+\tan^{-1}\left(\frac{a^2\tan x}{b^2}\right)\right)+C$

Q.9 $2\ln\frac{t}{2t+1}+\frac{1}{2t+1}+C$ when $t=x+\sqrt{x^2+x}$

Q.10 $\frac{1}{2}\ln(x^2+x+1)-\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x-1}{\sqrt{3}}\right)-\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x^2+1}{\sqrt{3}}\right)+C$

Q.11 $\cos a\cdot\arccos\left(\frac{\cos x}{\cos a}\right)-\sin a\cdot\ln\left(\sin x+\sqrt{\sin^2 x-\sin^2 a}\right)+C$

Q.12 $-\frac{3\left(1+4\tan^2 x\right)}{8(\tan x)^{8/3}}+C$

Q.13 $\frac{1}{2}\ln\left|\tan\frac{x}{2}\right|+\frac{1}{4}\sec^2\frac{x}{2}+\tan\frac{x}{2}+C$

Q.14 $(a+x)\arctan\sqrt{\frac{x}{a}}-\sqrt{ax}+C$

Q.15 $\frac{\left(x^2+1\right)\sqrt{x^2+1}}{9x^3}\cdot\left[2-3\ln\left(1+\frac{1}{x^2}\right)\right]$

Q.16 $\ln\left(\frac{xe^x}{1+xe^x}\right)+\frac{1}{1+xe^x}+C$

Q.17 3

Q.18 $-\ln(1-x^4)+C$

Q.19 $6\left[\frac{t^4}{4}-\frac{t^2}{2}+t+\frac{1}{2}\ln(1+t^2)-\tan^{-1}t\right]+C$ where $t=x^{1/6}$

Q.20 $\frac{4}{\sqrt{\cos\frac{x}{2}}}+2\tan^{-1}\sqrt{\cos\frac{x}{2}}-\ln\frac{1+\sqrt{\cos\frac{x}{2}}}{1-\sqrt{\cos\frac{x}{2}}}+C$

Q.21 $C-\ln(1+(x+1)e^{-x})-\frac{1}{1+(x+1)e^{-x}}$

Q.22 $\sin^{-1}\left(\frac{1}{2}\sec^2\frac{x}{2}\right)+C$

Q.23 $\frac{1}{24}\ln\frac{(4+3\sin x+3\cos x)}{(4-3\sin x-3\cos x)}+C$

Q.24 $\frac{1}{2}\left[\sin x-\cos x-\frac{1}{\sqrt{2}}\ln\tan\left(\frac{x}{2}+\frac{\pi}{8}\right)\right]+C$

Q.25 $\frac{1}{2\sqrt{3}}\ln\frac{\sqrt{3}+\sin x-\cos x}{\sqrt{3}-\sin x+\cos x}+\arctan(\sin x+\cos x)+C$

Q.26 $\left[-\ell n(\sec x)-\frac{1}{2}\ell n(\sec 2x)+\frac{1}{3}\ell n(\sec 3x)\right]+C$ **Q.27** $C-\frac{2}{\sin\alpha}\sqrt{\frac{\sin(x+\alpha)}{\sin x}}$

Q.28 $ln\left|\frac{x\sin x+\cos x}{x\cos x-\sin x}\right|$ **Q.29** $2x-3\arctan\left(\tan\frac{x}{2}+1\right)+C$

Q.30 $\frac{x^4}{4}+x^3-x^2+5x+\frac{1}{2}ln(x^2+1)+3\tan^{-1}x+C$

Q.31 $C-\frac{\sqrt{1+t^2}}{2}-\frac{1}{4}ln\frac{\sqrt{t^2+1}-1}{\sqrt{t^2+1}+1}$, where $t=\cot^2 x$ **Q.32** $C-\frac{x}{(x^2-1)^2}$

Q.33 $C-e^{\cos x}(x+\operatorname{cosec} x)$ **Q.34** $\sin^{-1}\left(\frac{ax^2+b}{cx}\right)+k$ **Q.35** $e^x\sqrt{\frac{1+x}{1-x}}+c$

Q.36 $\frac{2(7x-20)}{9\sqrt{7x-10-x^2}}+C$ **Q.37** $\operatorname{arcsec} x-\frac{\ln x}{\sqrt{x^2-1}}+C$

Q.38 $\sqrt{3}ln\frac{t-\sqrt{3}}{t+\sqrt{3}}+2\tan^{-1}(t)+C$ where $t=\sqrt{\frac{2-\sin x}{2+\sin x}}$

Q.39 $\tan^{-1}\left(\frac{\sqrt{2\sin 2x}}{\sin x+\cos x}\right)+C$ **Q.40** $4ln\,x+\frac{7}{x}+6\tan^{-1}(x)+\frac{6x}{1+x^2}+C$

Q.41 $\frac{-2}{\alpha-\beta}\sqrt{\frac{x-\beta}{x-\alpha}}+C$ **Q.42** $\frac{2}{3}\tan^{-1}(\sin x+\cos x)+\frac{1}{3\sqrt{2}}ln\left|\frac{\sqrt{2}+\sin x+\cos x}{\sqrt{2}-\sin x-\cos x}\right|+C$

Q.43 $\frac{1}{\sqrt{2}}ln\left(\frac{\sqrt{2}+t}{\sqrt{2}-t}\right)-\frac{1}{2}ln\left(\frac{1-t}{1+t}\right)$ where $t=\cos\theta$ and $\theta=\operatorname{cosec}^{-1}(\cot x)$

Q.44 $\frac{1}{2}\left(\operatorname{cosec}\frac{\alpha}{2}\right)\cdot\tan^{-1}\left(\left(\frac{x^2-1}{2x}\right)\operatorname{cosec}\frac{\alpha}{2}\right)$ **Q.45** $\frac{1}{2-\tan x}-\frac{1}{5}x+\frac{2}{5}\ln\sec x-\frac{2}{5}\ln|2-\tan x|+C$

Q.46 $2\tan^{-1}\sqrt{\frac{x^2+x+1}{x}}+\frac{2\sqrt{x}}{\sqrt{x^2+x+1}}+C$ **Q.47** $\ln|\tan^{-1}(\sec x+\cos x)|+C$

Q.48 $\ln\left|\frac{\sqrt{2xe^{\sin x}+1}-1}{\sqrt{2xe^{\sin x}+1}+1}\right|+C$ **Q.49** $\frac{12}{7}(1+x^{1/4})^{7/3}-3(1+x^{1/4})^{4/3}+C$

Q.50 (A) S; (B) P ; (C) Q ; (D) R

EXERCISE–II

Q.1 $\frac{\pi^2}{8}-\frac{\pi}{4}(1+l\text{n}2)+\frac{1}{2}$ **Q.3** **(a)** $I_n = e - n\,I_{n-1}$, $I_3 = 6 - 2e$; **(b)** n = 3

Q.4 $\frac{\pi}{2}-1$ **Q.6** 125 **Q.7** $\frac{1}{4\sqrt{2}}\ln(\sqrt{2}+1)+\frac{1}{4}$ **Q.8** 2525

Q.9 5250 **Q.10** 4 **Q.11** $l\text{n}\,2$ **Q.12** $4\sqrt{2}-4\,l\text{n}\,(\sqrt{2}+1)$

Q.13 $\frac{\pi}{2\sqrt{2}}-\frac{16\sqrt{2}}{5}$ **Q.14** $\frac{\pi}{8}l\text{n}\,2$ **Q.15** $\frac{\pi^2}{6\sqrt{3}}$ **Q.16** $\frac{\pi}{8}l\text{n}\,2$

Q.17 2008 **Q.18** $2\sqrt{6}$ **Q.19** 153 **Q.20** $\frac{\pi}{3}$

Q.21 $\frac{\pi(a+b)}{2\sqrt{2}}$ **Q.22** $\frac{\pi}{8}(1-l\text{n}\,4)$ **Q.23** 2021 **Q.24** $\frac{\pi\sqrt{3}}{3}$ **Q.25** $\frac{(a\pi+2b)\pi}{3\sqrt{3}}$

Q.26 $\frac{\pi(\pi+3)}{2}$ **Q.27** $\frac{5\pi}{27}$ **Q.28** $\frac{16\pi}{3}-2\sqrt{3}$ **Q.29** $\frac{2\pi}{\sqrt{3}}$

Q.30 $\tan^{-1}(a)\cdot l\text{n}\sqrt{1+a^2}$ **Q.31** $\frac{1}{2}\left[\frac{\pi}{6}+l\text{n}3-l\text{n}2\right]$ **Q.32** $-\frac{2\pi^2}{3}\,l\text{n}\,2$

Q.33 0 **Q.36** 10 **Q.37** $\frac{3\pi^2}{16}$ **Q.38** $\frac{\pi}{12}$ **Q.39** real & distinct $\forall$ $k \in R$

Q.40 $\frac{\pi+4}{666}$ **Q.41** $\frac{8}{\pi}$ **Q.42** **(b)** $\frac{\pi\theta}{4\sin\theta}$ **Q.43** $\frac{\pi}{16}$ **Q.45** I = 8 as $\int_0^{\pi/2} y\sin y\,dy = 1$

Q.47 $\frac{\pi}{2\sqrt{3}}$

EXERCISE–III

Q.2 $\left\{-\frac{\pi}{2},\frac{\pi}{2}\right\}$ **Q.3** cont. & der. at x = 0

Q.4 g(x) is cont. in (–2 , 2); g(x) is der. at x = 1 & not der. at x = 0 . Note that ;

$$g(x)=\begin{cases}-(x+2) & \text{for} \quad -2\le x\le 0\\ -2+x-\frac{x^2}{2} & \text{for} \quad 0<x<1\\ \frac{x^2}{2}-x-1 & \text{for} \quad 1\le x\le 2\end{cases}$$

Q.5 – cos x **Q.7** 1 + e **Q.9** **(a)** c = 1 and $\lim_{x\to\infty}$ will be $\frac{\sqrt{3}}{2}$ **(b)** a = 4 and b = 1

Q.10 13.5 **Q.11** a = 15, b = $\frac{45}{2}$ **Q.12** x = 2 or 4 **Q.13** $\frac{3}{8}$

Q.15 $f(x) = e^{x+1}$ **Q.17** 0 **Q.19** **(a)** $2e^{(1/2)(\pi-4)}$; **(b)** $3 - \ln 4$

Q.20 **(a)** $\frac{1}{e}$; **(b)** 11 **Q.21** $f(x) = 1 + x^2$ **Q.22** for n = 1, I > 0, n = 2, I = 0, $n \geq 3$, I < 0

Q.25 $f(x) = 3 + 2e^{2x}$; $g(x) = 3 - 2e^{2x}$ **Q.26** $f(x) = x + \frac{61}{119}x + \frac{80}{119}x^2$ **Q.28** $U_n = \frac{n\pi}{2}$

Q.29 **(a)** $(0, \infty)$; **(b)** $6 \ln 2$; **(c)** 126

EXERCISE–IV

1.	A	**2.**	A	**3.**	C	**4.**	D	**5.**	C	**6.**	B	**7.**	A
8.	B	**9.**	D	**10.**	D	**11.**	A	**12.**	C	**13.**	A	**14.**	A
15.	D	**16.**	D	**17.**	B	**18.**	D	**19.**	B	**20.**	B	**21.**	A
22.	C												

EXERCISE–V

1.	A, C	**2.**	A, B	**3.**	D	**4.**	A, B, C	**5.**	C, D
6.	9	**7.**	7	**8.**	A	**9.**	B,C	**10.**	1
11	B, C	**12.**	Bonus	**13.**	2	**14.**	AB	**15.**	1
16.	2	**17.**	4.00	**18.**	(A, B)	**19.**	0.50	**20.**	ABD
21.	4.00	**22.**	A,B,C	**23.**	2.00	**24.**	1.50	**25.**	182
26.	C, D	**27.**	5.00	**28.**	B	**29.**	C	**30.**	C
31.	0.00	**32.**	0.24 TO 0.26	**33.**	5.00	**34.**	21.00		